1.Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

knsat@SRITHIKA: ~

GNU nano 8.7.1

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char cmd[100];
    printf("Enter Linux command: ");
    scanf("%s", cmd);

    pid_t pid = fork();

    if (pid == 0) {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
    } else if (pid > 0) {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    return 0;
}
```

```
knsat@SRITHIKA:~$  nano exec_command.c
knsat@SRITHIKA:~$ gcc exec_command.c -o exec_command
knsat@SRITHIKA:~$ ./exec_command
Enter Linux command: uname
Parent PID : 1991
Child PID : 1992
Linux
Child process completed.
```

```
knsat@SRITHIKA: ~
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   8
  On-line CPU(s) list:    0-7
Vendor ID:                GenuineIntel
  Model name:             11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz
    CPU family:           6
    Model:                140
    Thread(s) per core:   2
    Core(s) per socket:   4
    Socket(s):            1
    Stepping:             1
    BogoMIPS:             5606.40
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht
syscall nx pdpe
                          1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known
_freq pni pclmu
                          lqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16
c rdrand hyperv
                          isor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase ts
c_adjust bmi1 a
                          vx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni a
vx512bw avx512v
                          l xsaveopt xsavec xgetbv1 xsaves vnmi avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx5
12_bitalg avx51
                          2_vpopcntdq rdpid movdiri movdir64b fsrm avx512_vp2intersect md_clear ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
```

```
knsat@SRITHIKA: ~
Caches (sum of all):
  L1d:                    192 KiB (4 instances)
  L1i:                    128 KiB (4 instances)
  L2:                     5 MiB (4 instances)
  L3:                     12 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-7
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Not affected
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI SW loop, KVM SW lo
op
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
knsat@SRITHIKA:~$
```

```
knsat@SRITHIKA:~$ ./exec_command
Enter Linux command: lsblk
Parent PID : 2092
Child PID : 2093
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                                /
Child process completed.
```

```
knsat@SRITHIKA:~$ ./exec_command
Enter Linux command: ps
Parent PID : 2105
Child PID : 2106
    PID TTY          TIME CMD
    745 pts/0    00:00:00 bash
   2105 pts/0    00:00:00 exec_command
   2106 pts/0    00:00:00 ps
Child process completed.
```

```
knsat@SRITHIKA: ~

top - 05:20:50 up  3:44,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  25 total,   1 running,  24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  1.1 sy,  0.0 ni, 97.9 id,  1.1 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7763.6 total,   6981.4 free,    530.9 used,    405.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7232.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24180  15488  11664 S   0.0   0.2   0:02.80 systemd
      2 root      20   0    3180   2212   2072 S   0.0   0.0   0:00.15 init-systemd(Ub
      7 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     47 root      19  -1   50428  17012  15716 S   0.0   0.2   0:01.66 systemd-journal
     79 systemd+  20   0   22416  14532  12040 S   0.0   0.2   0:00.27 systemd-resolve
     86 root      20   0   35328  12344   9300 S   0.0   0.2   0:05.36 systemd-udevd
    105 root      20   0    2888   2012   1884 S   0.0   0.0   0:00.07 chronyd-starter
    106 root      20   0    4472   3192   2944 S   0.0   0.0   0:00.03 cron
    107 message+  20   0    8812   5452   4780 S   0.0   0.1   0:00.36 dbus-daemon
    135 root      20   0   44096  29904  14760 S   0.0   0.4   0:00.28 networkd-dispat
    156 root      20   0   18432   9472   8236 S   0.0   0.1   0:00.35 systemd-logind
    182 syslog    20   0  220548   5804   4492 S   0.0   0.1   0:00.58 rsyslogd
    233 root      20   0    5492   2832   2652 S   0.0   0.0   0:00.00 agetty
    254 _chrony   20   0   23868  11448   7256 S   0.0   0.1   0:00.83 chronyd
    256 root      20   0  123460  32604  17988 S   0.0   0.4   0:00.17 unattended-upgr
    265 _chrony   20   0   12096   2528   1884 S   0.0   0.0   0:00.03 chronyd
    328 root      20   0    8644   5604   4764 S   0.0   0.1   0:00.00 login
    371 knsat     20   0   22336  13644  11028 S   0.0   0.2   0:00.17 systemd
    373 knsat     20   0   22916   4068   2064 S   0.0   0.1   0:00.00 (sd-pam)
    399 knsat     20   0    6136   5480   3928 S   0.0   0.1   0:00.03 bash
    737 root      20   0    3184   1124    980 S   0.0   0.0   0:00.00 SessionLeader
    738 root      20   0    3200   1260   1108 S   0.0   0.0   0:01.07 Relay(745)
    745 knsat     20   0    6268   5668   3996 S   0.0   0.1   0:00.20 bash
   2175 root      20   0 1866288  15280  12620 S   0.0   0.2   0:00.11 wsl-pro-service
   2219 knsat     20   0    8156   5676   3596 R   0.0   0.1   0:00.01 top
```